

Lab 07 - Expected verification evidence

Run from the lab directory after the rollback in section 6b has completed:

```
bash verification/checks.sh
```

Expected transcript

```
Lab 07 verification - namespace kcna-lab07

== Namespace and dataset
[PASS] namespace kcna-lab07 is Active (= Active)
[PASS] release plan read: r24.2 wants replicas=4 surge=1 unavailable=0 (ceiling 5 / floor 4)

== Deployment shape matches the release plan
[PASS] strategy type (= RollingUpdate)
[PASS] spec.replicas (= 4)
[PASS] maxSurge (= 1)
[PASS] maxUnavailable (= 0)
[PASS] revisionHistoryLimit (= 3)

== Rollback landed on the good release (r24.2)
[PASS] current image (= nginx:1.27-alpine)
[PASS] template release label (= r24.2)
[PASS] mounted ConfigMap (= portal-site-r24-2)

== Availability
[PASS] readyReplicas (= 4)
[PASS] availableReplicas (= 4)
[PASS] updatedReplicas (= 4)
[PASS] Progressing condition reason (= NewReplicaSetAvailable)

== Served content comes from data/site-r24.2.html
[PASS] page served by a live Pod (= RELEASE=r24.2)

== ReplicaSet generations retained
[PASS] between 2 and 4 ReplicaSets retained under revisionHistoryLimit=3 (found 3)
[PASS] exactly one ReplicaSet is scaled above zero (= 1)

== Service front door
[PASS] Service type (= ClusterIP)
[PASS] EndpointSlice backend addresses (= 4)

-----
Lab 07: 19 passed, 0 failed
ALL CHECKS PASSED
```

```
echo $?
```

```
0
```

Supporting evidence the learner should be able to produce

1. The rollout obeyed its ceiling and floor

Captured from the `--watch` in Step 4. TOTAL never exceeded `replicas + maxSurge = 5`; AVAILABLE never fell below `replicas - maxUnavailable = 4`.

DESIRED	UPDATED	TOTAL	AVAILABLE
4	4	4	4
4	1	5	4
4	2	5	4
4	3	5	4
4	4	5	4
4	4	4	4

2. The wedged rollout stayed available

```
kubectl -n kcna-lab07 get deploy freight-portal \
-o jsonpath='{range .status.conditions[*]}{.type}={.status} reason={.reason}{"\n"}{end}'
```

```
Available=True reason=MinimumReplicasAvailable
Progressing=False reason=ProgressDeadlineExceeded
```

with

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
freight-portal	4/4	1	4	9m

3. The real image-pull error

```
Warning Failed 41s (x4 over 2m11s) kubelet Failed to pull image "nginx:1.27-alpine-mfs-hotfix": failed to pull and unpack image "docker.io/library/nginx:1.27-alpine-mfs-hotfix": failed to resolve reference "docker.io/library/nginx:1.27-alpine-mfs-hotfix": docker.io/library/nginx:1.27-alpine-mfs-hotfix: not found
```

4. History after the rollback - revision 2 became revision 4

```
kubectl -n kcna-lab07 rollout history deployment/freight-portal
```

```
deployment.apps/freight-portal
REVISION  CHANGE-CAUSE
1          r24.1 - initial portal release
3          r24.3 - BROKEN image tag (deliberate)
4          r24.2 - customs banner, zero-unavailable rollout
```

5. The immutable-selector rejection

```
The Deployment "freight-portal" is invalid: spec.selector: Invalid value: v1.LabelSelector{MatchLabels:map[string]string{"app":"freight-portal", "tier":"frontend"}, MatchExpressions:[v1.LabelSelectorRequirement(nil)]}: field is immutable
```

Notes on variance

- * ReplicaSet name suffixes are pod-template-hash values and differ on every cluster - never assert on them.
- * Running checks.sh before the rollback (while revision 3 is wedged) will fail current image, template release label, updatedReplicas and Progressing condition reason. That is the intended signal, not a script defect.
- * RS retained accepts 2-4 because a learner who skips the r24.3 failure injection will have only two ReplicaSets.